The Indicator Constant on the L_p Scale

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2026-07-03

Status

This packet records a candidate partial result, likely valid. It answers one standard family inside item (9) of the “Looking Forward” section of Cwikel–Milman–Rochberg, arXiv:1404.2893. It does not characterize all Köthe spaces whose indicator constant is $\log 2$.

Source Context

The source note summarizes Kalton’s indicator formalism. For an indicator-like functional Φ, it writes

$$\Delta_\Phi(f,g) = \Phi(f) + \Phi(g) - \Phi(f+g),$$

and requires a constant $\delta(\Phi)$ such that

$$0 \le \Delta_\Phi(f,g) \le \delta(\Phi)\big(\|f\|_1 + \|g\|_1\big).$$

The note observes that for indicators of Köthe spaces one always has $\delta(\Phi) \le \log 2$, that equality holds for L^1, and then asks for what other spaces equality holds.

> perhaps more conveniently, with regard to Theorem 1.1 on p. 481, which is a finite dimensional “model” of Theorem 6.6 which Nigel formulates in his introduction to help prepare the reader for what is to follow. In connection with this question we also remark that the constant $\log 2$ plays a special role in these theorems. Furthermore, whenever Φ is the indicator of a Köthe space, the optimal (smallest) constant $\delta(\Phi)$ for which (2.5) holds satisfies $\delta(\Phi) \le \log 2$. Equality holds when the Köthe space is L^1. For what other spaces, if any, does equality hold?
>
> (10) As Nigel points out on p. 510, the centralizer $\Omega^{[1]}$ obtained when $\Omega = \Omega(A_0, A_1, t)$ is the same for all values of $t \in (0,1)$. Can we somehow interpret this to mean that, when A_0 and A_1 are both Köthe function spaces, the “geodesic curve” $\{[A_0, A_1]_t : 0 < t < 1\}$ which joins them is in some

Proof Intuition

For L_p, the optimizing vector in the indicator definition is forced by a one-line Lagrange multiplier calculation: x^p is proportional to f. Thus the L_p indicator is exactly $1/p$ times the L_1 entropy indicator. The defect Δ_Φ is therefore also scaled by $1/p$. Kalton’s universal $\log 2$ bound is just the binary entropy bound, and disjoint equal-mass functions make this bound sharp.

Result

Let (S, Σ, μ) be a measure space. For $f \geq 0$, set

$$\Lambda(f) = \int_S f \log f \, d\mu - \|f\|_1 \log \|f\|_1,$$

with the usual conventions $0 \log 0 = 0$ and $\Lambda(0) = 0$.

Theorem 1 (Exact L_p indicator constants)**.** *Let $1 \leq p < \infty$. On the natural finite-entropy domain of the indicator for $L_p(S, \mu)$,*

$$\Phi_{L_p}(f) = \frac{1}{p} \Lambda(f).$$

Consequently,

$$0 \leq \Delta_{\Phi_{L_p}}(f, g) \leq \frac{\log 2}{p} \big(\|f\|_1 + \|g\|_1 \big).$$

If the measure algebra contains two disjoint sets of finite positive measure, then the optimal constant is exactly $(\log 2)/p$. For L_∞, the optimal constant is 0.

Proof. Fix $1 \leq p < \infty$ and $f \geq 0$, $f \neq 0$, with finite entropy. Writing $m = \|f\|_1$, the indicator is

$$\Phi_{L_p}(f) = \sup_{\|x\|_p \leq 1} \int_S f \log |x| \, d\mu.$$

The strict concavity of $x \mapsto \log x$ and the constraint $\int |x|^p \, d\mu \leq 1$ give the maximizer

$$x_f = \left(\frac{f}{m} \right)^{1/p}$$

on the support of f. Therefore

$$\Phi_{L_p}(f) = \frac{1}{p} \int_S f \log \frac{f}{m} \, d\mu = \frac{1}{p} \Lambda(f).$$

The case $f = 0$ is immediate.

It remains to recall the sharp entropy-defect bound for Λ. Let $a = \|f\|_1$, $b = \|g\|_1$, $h = f + g$, and $m = a + b$. On the set where $h > 0$, put $u = f/h$, and define the binary entropy

$$H(t) = -t \log t - (1 - t) \log(1 - t).$$

A direct expansion gives

$$\Delta_\Lambda(f, g) = mH(a/m) - \int_S h \, H(u) \, d\mu.$$

Since H is concave and $\int hu \, d\mu = a$, Jensen's inequality gives $\int hH(u) \, d\mu \leq mH(a/m)$. Hence $\Delta_\Lambda(f, g) \geq 0$. Also $H(a/m) \leq \log 2$, so

$$\Delta_\Lambda(f, g) \leq m \log 2.$$

Multiplying by $1/p$ proves the displayed upper bound.

If E, F are disjoint measurable sets with $0 < \mu(E), \mu(F) < \infty$, take

$$f = \mu(E)^{-1} \mathbf{1}_E, \qquad g = \mu(F)^{-1} \mathbf{1}_F.$$

Then $a = b = 1$, $u \in \{0, 1\}$ almost everywhere on the support of h, so $\int hH(u) = 0$ and $\Delta_\Lambda(f, g) = 2 \log 2$. The ratio $\Delta_{\Phi_{L_p}}(f, g)/(\|f\|_1 + \|g\|_1)$ is therefore $(\log 2)/p$, proving optimality.

For L_∞, the constraint $\|x\|_\infty \leq 1$ gives $\log |x| \leq 0$, while $x = 1$ gives value 0. Thus $\Phi_{L_\infty} = 0$ and the optimal defect constant is 0. $\square$

Corollary 1 (Weighted L_p spaces)**.** *Let $w > 0$ a.e. and $1 \le p < \infty$. For $L_p(w\,d\mu)$,*

$$\Phi_{L_p(w)}(f) = \frac{1}{p}\Lambda(f) - \frac{1}{p}\int_S f \log w \, d\mu$$

whenever the right-hand side is finite. Hence the same defect bound and the same sharp constant $(\log 2)/p$ hold whenever the measure space admits the disjoint equal-mass test functions in the indicator domain.

Proof. The constrained maximizer is now

$$x_f = \left(\frac{f}{\|f\|_1 w}\right)^{1/p}.$$

Substitution gives the displayed formula. The additional weight term is linear in f, so it cancels from $\Phi(f) + \Phi(g) - \Phi(f+g)$. The rest is Theorem 1. $\square$

Implication for the Source Question

The L_p scale does not provide new equality cases beyond the L^1 endpoint. In every nondegenerate measure space covered by the theorem,

$$\delta(\Phi_{L_p}) = \frac{\log 2}{p},$$

so equality with Kalton's universal upper bound $\log 2$ occurs exactly at $p = 1$ in this scale. Weighted L_1 spaces have the same extremal constant, as expected from the cancellation of linear weight terms.

Limitations and Novelty Check

This is only a partial result. It does not rule out non-L_1 Köthe spaces with $\delta(\Phi) = \log 2$, and it does not address the limiting-case questions about Kalton's Theorems 1.1 and 6.6 that immediately precede the indicator-constant question in the source.

Local run indexes were searched for arXiv:1404.2893, Kalton, Köthe, centralizer, indicator, $\delta(\Phi)$, and $\log 2$. No duplicate packet for this exact calculation was found. A bounded web search for the exact phrases around "delta(Phi)", "log 2", "Kothe", and "indicator" found the source paper but no exact preexisting answer to this subcase. The calculation is elementary and may be folklore; the packet records it because it gives a clean, reviewable partial answer to item (9).

Human Review Recommendation

Likely valid. Review should check the conventions in Kalton's indicator domain and the nondegeneracy assumptions needed for sharpness on the chosen measure space. The proof of the formula and the upper bound is independent of those sharpness assumptions.

References

[1] M. Cwikel, M. Milman, and R. Rochberg, *An introduction to Nigel Kalton's work on differentials of complex interpolation processes for Kothe spaces*, arXiv:1404.2893, 2014.

[2] N. J. Kalton, *Differentials of complex interpolation processes for Kothe function spaces*, Trans. Amer. Math. Soc. 333 (1992), no. 2, 479–529.